# Software Requirements Specification

### AI-Driven Surveillance Drone Using Agile Software Engineering

# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the software and hardware requirements for the "AI-Driven Surveillance Drone Using Agile Software Engineering" project. This document serves as the primary technical baseline for the Major-I phase, ensuring that all stakeholders, including developers and project evaluators, have a clear understanding of the hexacopter system architecture, companion computing integration, and surveillance objectives.

## 1.2 Scope

The scope of this Major-I project encompasses the development of an S550 Hexacopter platform. This includes the mechanical assembly, integration of the Pixhawk 2.4.8 flight controller with ArduPilot firmware, and the implementation of a companion computing system using a Raspberry Pi 5 (16GB) and Pi Camera. The project focuses on establishing robust telemetry, real-time data visualization via a dashboard, and the foundational software architecture for AI-driven computer vision and surveillance.

## 1.3 Definitions, Acronyms and Abbreviations

- **AI:** Artificial Intelligence.
- **SRS:** Software Requirements Specification.
- **ESC:** Electronic Speed Controller.
- **GPS:** Global Positioning System.
- **RC:** Remote Control.
- **MAVLink:** Micro Air Vehicle Link communication protocol.
- **LiPo:** Lithium Polymer Battery.
- **CW / CCW:** Clockwise / Counter-Clockwise (motor rotation).
- **ArduPilot:** Open-source autopilot software system.
- **Telemetry:** The wireless transmission of data from the drone to a ground station.
- **Raspberry Pi:** A high-performance companion computer for vision processing.

- **Pixhawk:** An advanced flight controller hardware platform.
- **Agile:** An iterative approach to software development.
- **Sprint:** A time-boxed period in Agile during which specific work is completed.

### 1.4 References


1. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
2. ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*.
3. ArduPilot Documentation: *Copter Mission Planning and Hardware Integration*.


### 1.5 Document Overview

This document is organized into eleven sections. Section 2 describes the product perspective and high-level functions. Section 3 details specific functional requirements. Section 4 defines interfaces, while Section 5 covers non-functional attributes. Section 7 explicitly details the Agile methodology applied to this development cycle.

## 2. Overall Description

### 2.1 Product Perspective

The AI-Driven Surveillance Drone is an integrated system comprising an S550 Hexacopter airframe, a Pixhawk flight control subsystem, and a Raspberry Pi-based companion computer. Unlike standard drones, this system utilizes a parallel computing architecture where flight stabilization is handled by the Pixhawk, while high-level AI and vision tasks are offloaded to the Raspberry Pi 5. Data from both subsystems is aggregated and transmitted via 433MHz telemetry to a real-time monitoring dashboard.

### 2.2 Product Functions

- **Flight Stabilization:** Maintains multi-rotor stability using six brushless motors and PID tuning.
- **Navigation:** Provides real-time GPS positioning and heading via compass.
- **Vision Acquisition:** Captures high-definition visual data through the Pi Camera.
- **Companion Processing:** Executes computer vision algorithms on the Raspberry Pi 5.
- **Telemetry Link:** Transmits flight parameters and RPi status to the ground station.
- **Ground Monitoring:** Displays a real-time dashboard of all system vitals and vision results.

## 2.3 User Classes and Characteristics

- **Drone Operator/Pilot:** Requires manual control via RC and visual monitoring via the dashboard.
- **Project Administrator/Developer:** Responsible for firmware configuration, code deployment, and system calibration.
- **Ground Station Operator:** Monitors live telemetry data and surveillance feeds for reporting.

## 2.4 Operating Environment

- **Outdoor Environment:** Variable wind conditions and open sky for GPS signal acquisition.
- **Onboard Software:** ArduPilot Copter firmware and Linux-based OS on Raspberry Pi.
- **Ground Environment:** Laptop or desktop running dashboard software and telemetry receiver.

## 2.5 Design and Implementation Constraints

- **Configuration:** Restricted to an S550 Hexacopter (6-motor) platform.
- **Power:** 11.1V 8000mAh 3S LiPo battery limits total flight duration.
- **Telemetry:** 433MHz frequency restricts high-bandwidth data transmission; primary use is for MAVLink telemetry.
- **Weight:** Component layout must respect the hexacopter's center of gravity and maximum lift capacity.
- **Thermal:** Raspberry Pi 5 requires adequate cooling for continuous AI processing.

## 2.6 Assumptions and Dependencies

- **Hardware Availability:** All specified components (Pixhawk, RPi 5, S550 frame) are available for assembly.
- **GPS Reliability:** Outdoor testing assumes a minimum of 6 satellites for stable Loiter/Auto modes.
- **Communication:** 2.4GHz RC and 433MHz telemetry links are assumed to be free from significant local interference.

# 3. System Features and Functional Requirements

## 3.1 Hexacopter Assembly and Hardware Integration

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-001 | The system shall integrate six brushless motors onto the S550 hexacopter frame in the designated CW/CCW pattern. | High | Inspection |
| FR-002 | The system shall utilize a Power Module V6.0 to distribute power from the 11.1V LiPo battery to the ESCs and Pixhawk. | High | Functional Test |

## 3.2 Flight Control System

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-003 | The system shall use the Pixhawk 2.4.8 running ArduPilot firmware for flight stabilization and control. | High | Software Verification |
| FR-004 | The system shall support manual | High | Functional Test |

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| | flight control via the FS-i6S transmitter and FS-IA10B receiver. | | |

### 3.3 Motor and ESC Control

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-005 | The system shall utilize six 40A ESCs to regulate motor speed based on PWM signals from the Pixhawk. | High | Functional Test |
| FR-006 | The system shall provide a manual arming/disarming sequence for motor safety. | High | Functional Test |

### 3.4 GPS and Navigation Data

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-007 | The system shall integrate the NEO-M8N GPS + Compass module for location and | High | Inspection |

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| | heading acquisition. | | |
| FR-008 | The system shall provide real-time latitude, longitude, and altitude data to the telemetry stream. | Medium | Data Analysis |

## 3.5 Pre-Flight Checks and Calibration

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-009 | The system shall undergo mandatory Accelerometer and Magnetometer calibration prior to flight. | High | Inspection |
| FR-010 | The system shall perform an ESC calibration to ensure synchronized motor startup. | High | Functional Test |

## 3.6 First Flight Testing

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-011 | The system shall undergo a controlled hover test to verify flight stability. | High | Flight Testing |
| FR-012 | The system shall demonstrate basic movement (Pitch, Roll, Yaw) under pilot command. | High | Flight Testing |

## 3.7 Raspberry Pi 5 Integration

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-013 | The system shall mount a Raspberry Pi 5 (16GB) as a companion computer for high-level tasks. | High | Inspection |
| FR-014 | The Raspberry Pi 5 shall be powered via a dedicated 5V supply from the power distribution system. | Medium | Functional Test |

## 3.8 Pi Camera 5MP Integration

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-015 | The system shall integrate a 5MP Pi Camera connected to the Raspberry Pi 5 via the CSI interface. | High | Inspection |
| FR-016 | The system shall allow for the capture of real-time images or video streams for surveillance processing. | Medium | Functional Test |

## 3.9 AI-Based Surveillance and Vision Processing

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-017 | The system shall utilize the Raspberry Pi 5 to process visual data for basic anomaly or object detection. | Medium | Software Test |
| FR-018 | [Planned Requirement] The vision system shall output processed results to the | Low | Integration Test |

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| | telemetry or dashboard. | | |

## 3.10 Raspberry Pi and Pixhawk Communication

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-019 | The system shall establish a serial communication link (MAVLink) between the Pixhawk and Raspberry Pi 5. | High | Communication Test |
| FR-020 | The Raspberry Pi shall be able to request and receive flight telemetry data from the Pixhawk. | Medium | Software Verification |

## 3.11 Telemetry Data Communication

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-021 | The system shall use a 433MHz Telemetry Kit to transmit air-to-ground data. | High | Functional Test |

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-022 | The telemetry link shall support a range sufficient for line-of-sight monitoring. | Medium | Flight Testing |

## 3.12 Sensor and Flight Data Collection

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-023 | The system shall collect battery voltage, current draw, and flight mode status for ground transmission. | High | Data Analysis |
| FR-024 | The system shall aggregate Raspberry Pi health data (CPU usage, Temperature) into the telemetry stream. | Medium | Software Test |

### 3.13 Real-Time Dashboard

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-025 | The system shall provide a graphical dashboard to display GPS, battery, and flight mode data. | High | Functional Test |
| FR-026 | The dashboard shall visualize Raspberry Pi processing status and vision alerts. | Medium | GUI Inspection |

### 3.14 Data Logging

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-027 | The system shall log flight data to the Pixhawk's onboard SD card for post-flight analysis. | Medium | Inspection |

### 3.15 System Status, Alerts and Notifications

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| FR-028 | The system shall notify the pilot of | High | Functional Test |

| Requirement ID | Requirement Description | Priority | Verification Method |
|---|---|---|---|
| | critical battery levels via the GCS or Dashboard. | | |

# 4. External Interface Requirements

## 4.1 User Interfaces

The primary user interface is the **Real-Time Dashboard**. It shall provide a consolidated view of the following available data:

- **Flight Information:** Altitude, Speed, GPS Location (Map view), and Flight Mode.
- **Power Management:** Battery Voltage and Percentage.
- **Companion Diagnostics:** Raspberry Pi CPU usage, RAM usage, and Temperature.
- **Vision Data:** Live camera feed (where bandwidth permits) and AI detection alerts.
- **Connectivity:** Telemetry signal strength and heartbeat status.

## 4.2 Hardware Interfaces

| Component | Purpose | Input/Output | Connected Subsystem |
|---|---|---|---|
| Pixhawk 2.4.8 | Flight Control | I: Sensors, RC / O: ESCs | Main Air Unit |
| Raspberry Pi 5 | AI Companion | I: Camera, MAVLink / O: MAVLink | Parallel Computing |
| Pi Camera 5MP | Visual Input | I: Light / O: CSI Data | Raspberry Pi |
| NEO-M8N | Nav/Heading | I: Satellite Signal / O: UART | Pixhawk |
| Motors (6x) | Propulsion | I: Phase Current | S550 Frame |

| Component | Purpose | Input/Output | Connected Subsystem |
|---|---|---|---|
| 433MHz Kit | Ground Link | I: Serial / O: Radio Frequency | Telemetry System |
| Power Module | Energy Monitoring | I: LiPo / O: V/I Sensing | Pixhawk / Battery |

### 4.3 Software Interfaces

- **ArduPilot (Copter):** Core flight control logic.
- **MAVLink Protocol:** Standard for communication between Pixhawk, RPi, and Ground Station.
- **Linux (Raspberry Pi OS):** Operating environment for vision scripts.
- **OpenCV/TensorFlow Lite:** [Planned] Frameworks for vision processing.
- **Dashboard Framework:** (e.g., Python/Flask or dedicated GCS software) for visualization.

### 4.4 Communication Interfaces

- **RC Communication:** 2.4GHz link between FS-i6S and FS-IA10B.
- **Telemetry Communication:** 433MHz serial-to-radio link.
- **Inter-Processor Comm:** Physical UART/USB connection between Pixhawk and Raspberry Pi 5.

## 5. Non-Functional Requirements

| ID | Category | Requirement Description |
|---|---|---|
| NFR-001 | Performance | Telemetry data latency between drone and ground shall not exceed 300ms. |
| NFR-002 | Reliability | The system shall maintain a "Return to Launch" (RTL) capability for signal loss. |

| ID | Category | Requirement Description |
|---|---|---|
| NFR-003 | Safety | Propellers shall be secured with thread locker; all motors must be tested for rotation direction. |
| NFR-004 | Security | Access to the dashboard and flight control should be restricted to authorized devices. |
| NFR-005 | Usability | The dashboard interface shall display data in a readable format for outdoor use. |
| NFR-006 | Maintainability | The hexacopter modular frame design shall allow for easy replacement of arms or motors. |

# 6. System Constraints

- **SC-001:** The system must function as a **Hexacopter**; quadcopter configurations are not permitted.
- **SC-002:** Communication is limited to the bandwidth of the 433MHz telemetry kit for flight data.
- **SC-003:** Raspberry Pi processing is constrained by the thermal limits of the 16GB model during high-load AI tasks.
- **SC-004:** Total weight must not exceed the thrust-to-weight ratio (ideally 2:1) for safe flight operations.

# 7. Agile Software Engineering Approach

## 7.1 Agile Methodology Overview

This project follows an incremental development model. The system is broken down into modular sprints, allowing for continuous integration and testing of the propulsion, control, and vision layers.

## 7.2 Product Backlog (Prioritized)

1. Mechanical Assembly of S550 Frame.
2. Pixhawk/ArduPilot configuration.
3. Telemetry and RC integration.
4. Stability testing (First Flight).
5. Companion computer (RPi 5) setup.
6. Vision/Camera integration.
7. Dashboard development.
8. System-wide integration and AI optimization.

## 7.3 Sprint Planning

### Sprint 1: Drone Assembly and Hardware Integration

- **Goal:** Complete physical S550 Hexacopter structure.
- **Deliverables:** Assembled frame with 6 motors, 6 ESCs, and Power Module.
- **Criteria:** Motors must spin freely; structure must be rigid.

### Sprint 2: Pixhawk, ArduPilot, GPS and RC Configuration

- **Goal:** Establish flight control intelligence.
- **Deliverables:** Pixhawk with ArduPilot firmware and GPS lock.
- **Criteria:** Calibration successful; RC inputs correctly mapped in GCS.

### Sprint 3: Pre-Flight Testing and First Flight

- **Goal:** Verify airworthiness.
- **Deliverables:** Successful maiden hover test.
- **Criteria:** Stable hover in Stabilize/Alt-Hold modes.

### Sprint 4: Raspberry Pi 5 and Pi Camera Integration

- **Goal:** Implement companion computing hardware.

- **Deliverables:** RPi 5 mounted and Pi Camera streaming internally.
- **Criteria:** RPi boots successfully from drone power; camera captures image.

### Sprint 5: Raspberry Pi ↔ Pixhawk Communication and Vision Integration

- **Goal:** Enable inter-system data exchange.
- **Deliverables:** MAVLink heartbeat between RPi and Pixhawk.
- **Criteria:** RPi can read Pixhawk telemetry parameters.

### Sprint 6: Telemetry and Sensor/Flight Data Collection

- **Goal:** Establish air-to-ground data link.
- **Deliverables:** Continuous data stream to ground unit.
- **Criteria:** No packet loss within 10 meters distance.

### Sprint 7: Dashboard Development and Real-Time Monitoring

- **Goal:** Visualize system status.
- **Deliverables:** Functional GUI displaying drone telemetry and RPi vitals.
- **Criteria:** Real-time updates with minimal lag.

### Sprint 8: Integration and Documentation

- **Goal:** Final system validation.
- **Deliverables:** Complete project report and final tested system.
- **Criteria:** All Major-I objectives met.

## 8. Verification and Validation

| Requirement ID | Requirement | Module | Method | Status |
|---|---|---|---|---|
| FR-001 | Hexacopter Frame Assembly | Hardware | Inspection | Planned |
| FR-003 | Pixhawk Configuration | Control | Functional | Planned |
| FR-011 | Hover Test | Flight | Flight Test | Planned |

| Requirement ID | Requirement | Module | Method | Status |
|---|---|---|---|---|
| FR-013 | RPi 5 Integration | Companion | Integration | Planned |
| FR-021 | Telemetry Link | Comms | Comm Test | Planned |

# 9. Risk Analysis

| Risk ID | Risk | Impact | Probability | Mitigation |
|---|---|---|---|---|
| R-001 | Motor/ESC Failure | High | Low | Pre-flight stress test of each motor. |
| R-002 | Loss of RC Signal | High | Medium | Configure Failsafe: Return to Launch. |
| R-003 | GPS Glitch | Medium | Medium | Maintain manual control override at all times. |
| R-004 | RPi Overheating | Medium | High | Install active cooling (fan/heatsink) on RPi 5. |

# 10. Future Enhancements

- Autonomous mission planning via waypoints.
- Advanced AI-based object tracking and classification.
- Obstacle avoidance using LiDAR or Depth Cameras.
- Encryption of telemetry and vision data streams.

# 11. Appendices

## Appendix A: Glossary

- **Companion Computer:** A secondary processor (RPi) used for non-flight-critical tasks.
- **MAVLink:** Protocol used for communication between the autopilot and the companion computer.

## Appendix B: Drone Hardware Components

| Component | Model | Quantity |
|---|---|---|
| Frame | S550 Hexacopter | 1 |
| Motors | Brushless (CW/CCW) | 6 |
| ESCs | 40A | 6 |
| Flight Controller | Pixhawk 2.4.8 | 1 |
| Companion PC | Raspberry Pi 5 (16GB) | 1 |
| Camera | Pi Camera 5MP | 1 |
| Battery | 11.1V 8000mAh 3S LiPo | 1 |